

1-Finding Duplicates- $O(n^2)$ Time Complexity, $O(1)$ Space Complexity

QUESTION:

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[n];
```

```
    for(int i = 0; i < n; i++)
```

```
        scanf("%d", &arr[i]);
```

```
    for(int i = 0; i < n; i++) {
```

```
        for(int j = i + 1; j < n; j++) {
```

```
            if(arr[i] == arr[j]) {
```

```
                printf("%d", arr[i]);
```

```
                return 0;
```

```
            }
```

```
        }
```

```
    }
```

```

    return 0;
}

```

RESULT:

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

2-Finding Duplicates- $O(n)$ Time Complexity, $O(1)$ Space Complexity

QUESTION:

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```

scanf("%d", &n);

int arr[n];

for(int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
}

int slow = arr[0];
int fast = arr[0];
do {
    slow = arr[slow];
    fast = arr[arr[fast]];
} while(slow != fast);

slow = arr[0];
while(slow != fast) {
    slow = arr[slow];
    fast = arr[fast];
}

printf("%d", slow);

return 0;
}

```

RESULT:

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

3-Print Intersection of 2 sorted arrays- $O(m*n)$ Time Complexity, $O(1)$ Space Complexity

QUESTION:

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6	10 57

Input	Result
2 7 10 15 57 246	

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int T;
```

```
    scanf("%d", &T);
```

```
    while(T-->0) {
```

```
        int m, n;
```

```
        scanf("%d", &m);
```

```
        int a[m];
```

```
        for(int i = 0; i < m; i++) {
```

```
            scanf("%d", &a[i]);
```

```
        }
```

```
        scanf("%d", &n);
```

```
        int b[n];
```

```
        for(int i = 0; i < n; i++) {
```

```
            scanf("%d", &b[i]);
```

```
        }
```

```
        int first = 1;
```

```
        for(int i = 0; i < m; i++) {
```

```
            for(int j = 0; j < n; j++) {
```

```
                if(a[i] == b[j]) {
```

```
                    if(!first)
```

```
                        printf(" ");
```

```
                    printf("%d", a[i]);
```

```
                    first = 0;
```

```
                    break;
```

```
                }
```

```

    }

}

printf("\n");

}

return 0;

}

```

RESULT:

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

4-Print Intersection of 2 sorted arrays- $O(m+n)$ Time Complexity, $O(1)$ Space Complexity

QUESTION:

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int T;
```

```
    scanf("%d", &T);
```

```
    while(T-->0) {
```

```
        int m, n;
```

```
        scanf("%d", &m);
```

```
        int a[m];
```

```
        for(int i = 0; i < m; i++) {
```

```
            scanf("%d", &a[i]);
```

```
        }
```

```
        scanf("%d", &n);
```

```
        int b[n];
```

```
        for(int i = 0; i < n; i++) {
```

```
            scanf("%d", &b[i]);
```

```
        }
```

```
        int i = 0, j = 0;
```

```
        int first = 1;
```

```
        while(i < m && j < n) {
```

```
            if(a[i] == b[j]) {
```

```
                if(!first)
```

```
                    printf(" ");
```

```
                printf("%d", a[i]);
```

```
                first = 0;
```

```
                i++;
```

```
                j++;
```

```
            }
```

```
            else if(a[i] < b[j]) {
```

```
                i++;
```

```
            }
```

```

        else {
            j++;
        }
    }
    printf("\n");
}
return 0;
}

```

RESULT:

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

5-Pair with Difference- $O(n^2)$ Time Complexity, $O(1)$ Space Complexity

QUESTION:

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    for(int i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
    }
```

```
    int k;
```

```
    scanf("%d", &k);
```

```

for(int i = 0; i < n; i++) {
    for(int j = i + 1; j < n; j++) {
        if(a[j] - a[i] == k || a[i] - a[j] == k) {
            printf("1");
            return 0;
        }
    }
}
printf("0");
return 0;
}

```

RESULT:

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

QUESTION:

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

PROGRAM:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    for(int i = 0; i < n; i++) {
```

```
        scanf("%d", &a[i]);
```

```
    }
```

```

int k;

scanf("%d", &k);

int i = 0;

int j = 1;

while(j < n) {

    int diff = a[j] - a[i];

    if(i != j && diff == k) {

        printf("1");

        return 0;

    }

    else if(diff < k) {

        j++;

    }

    else {

        i++;

    }

    if(i == j)

        j++;

}

printf("0");

return 0;

}

```

RESULT:

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓